

Paired tumour-vs-normal T cells from 43 patients: which genes does a foundation model read as T-cell state?

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43 donors, both tissues,
100 cells each

Delete AND overexpress,
donor's own normal as goal

Registered before any number;
all 54 rows reported

Geneformer V2-316M (bf16) · in silico perturbation is a hypothesis generator, not a knockout screen

Two registered questions, one cleaner design

Panel A — do the July hits replicate?

The 15 LUAD→normal top genes of the July screen.
July compared LUAD with normal tissue from other studies (study-confounded).

Panel B — does the model read T-cell genes?

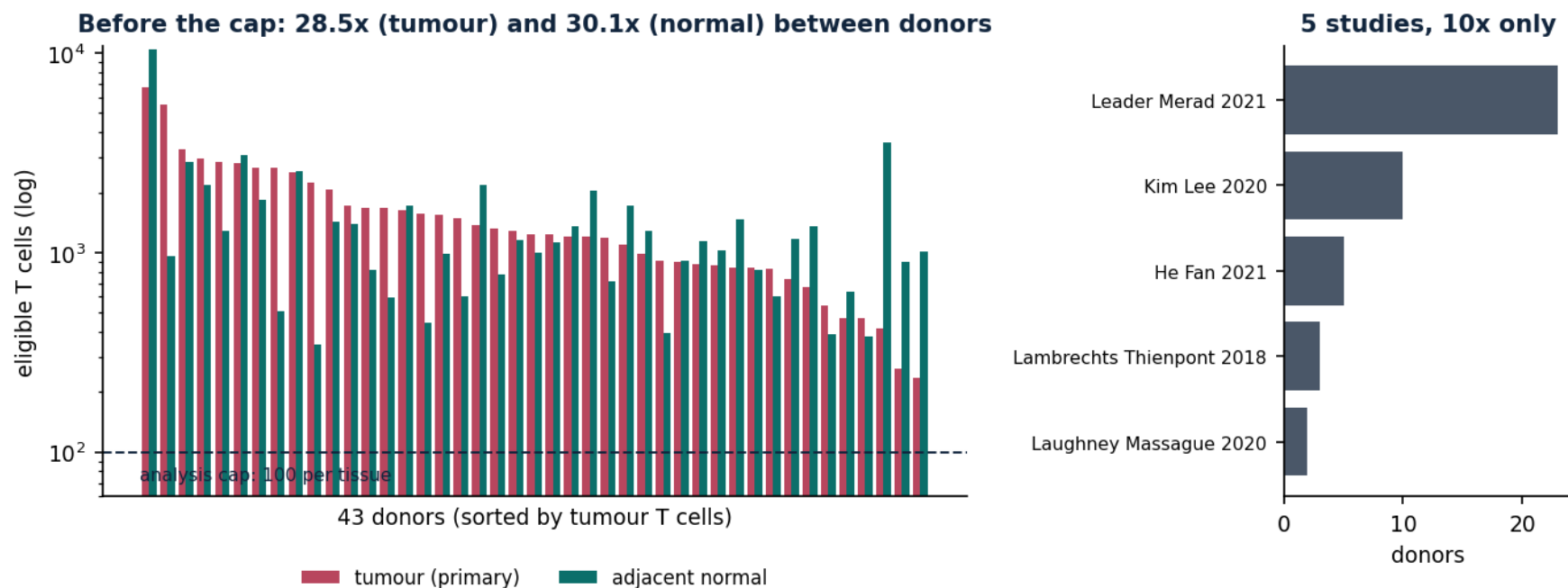
The lab's 39 curated T-cell genes (36 perturbable).
July never ranked them in its top 120.

What 'a hit' means here

Per donor: shift of tumour T cells toward the SAME donor's normal T cells, minus the median of 20 matched control genes.
Exact Wilcoxon over donors, Holm per panel, for deletion and overexpression separately.
A signal needs BOTH arms significant with opposite signs (dose concordance).

Pairing within a donor cancels study, chemistry and person — not ambient RNA

Design: every donor contributes both tissues; after the cap each donor weighs exactly 1/43

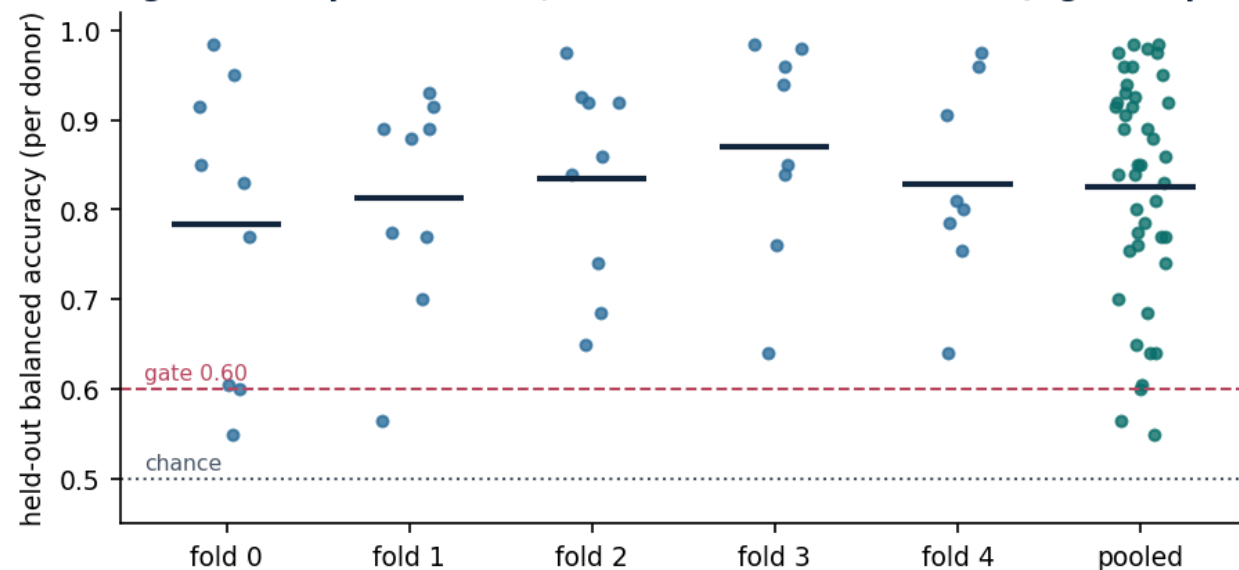


What pairing cannot fix

Tumour and normal tissue differ in ambient contamination within each donor. 20 of 43 donors overlap July's LUAD pool. Selection favours T-cell-rich samples; 17 unpaired donors excluded.

The model tells a donor's tumour from normal T cells: 43/43 donors above chance

Classifier gate PASS: pooled 0.825; 43/43 donors above chance (sign test $p = 2.3e-13$)



Gate: PASS

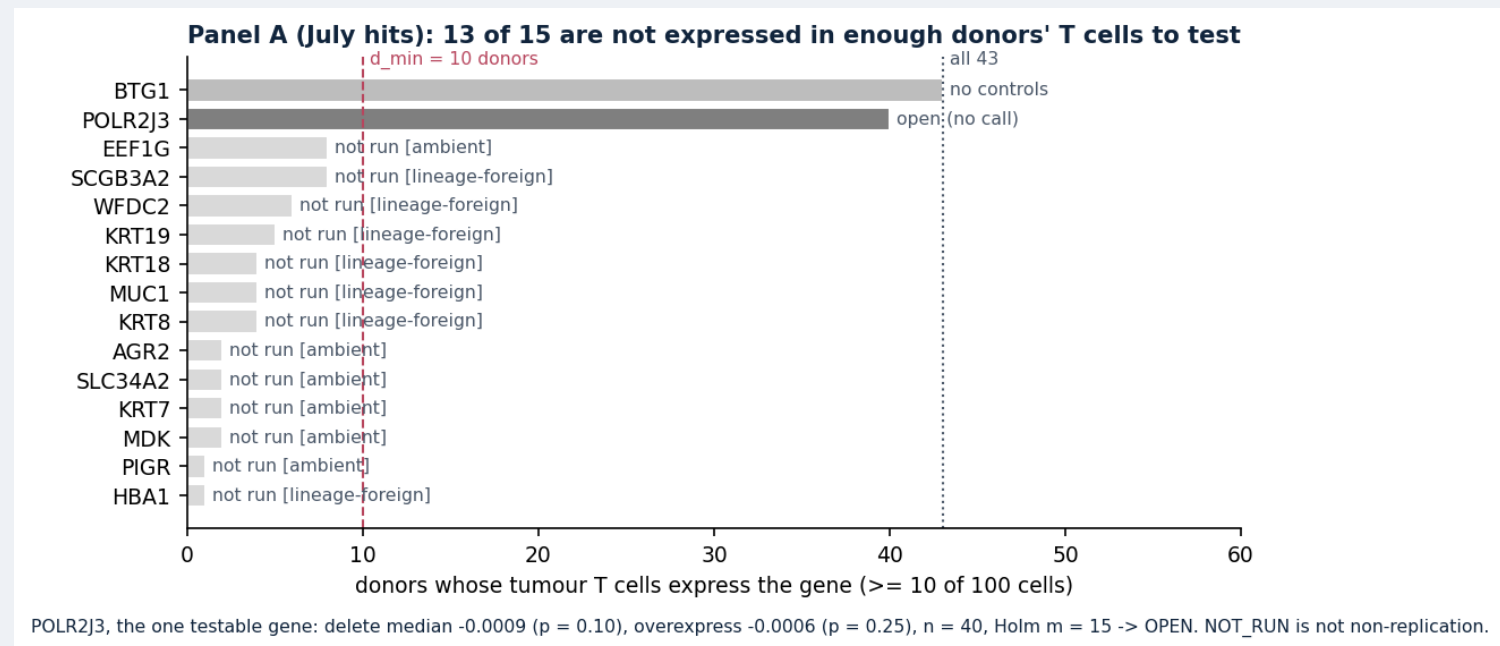
Pooled balanced accuracy 0.825 (gate 0.60).

Sign test $p = 2.3e-13$.

Every donor scored by a model that never saw it.

Fine-tune not bit-reproducible (band 0.045 per donor); cannot flip the gate.

13 of 15 July hits are not expressed in T cells — not testable, not refuted



Reading

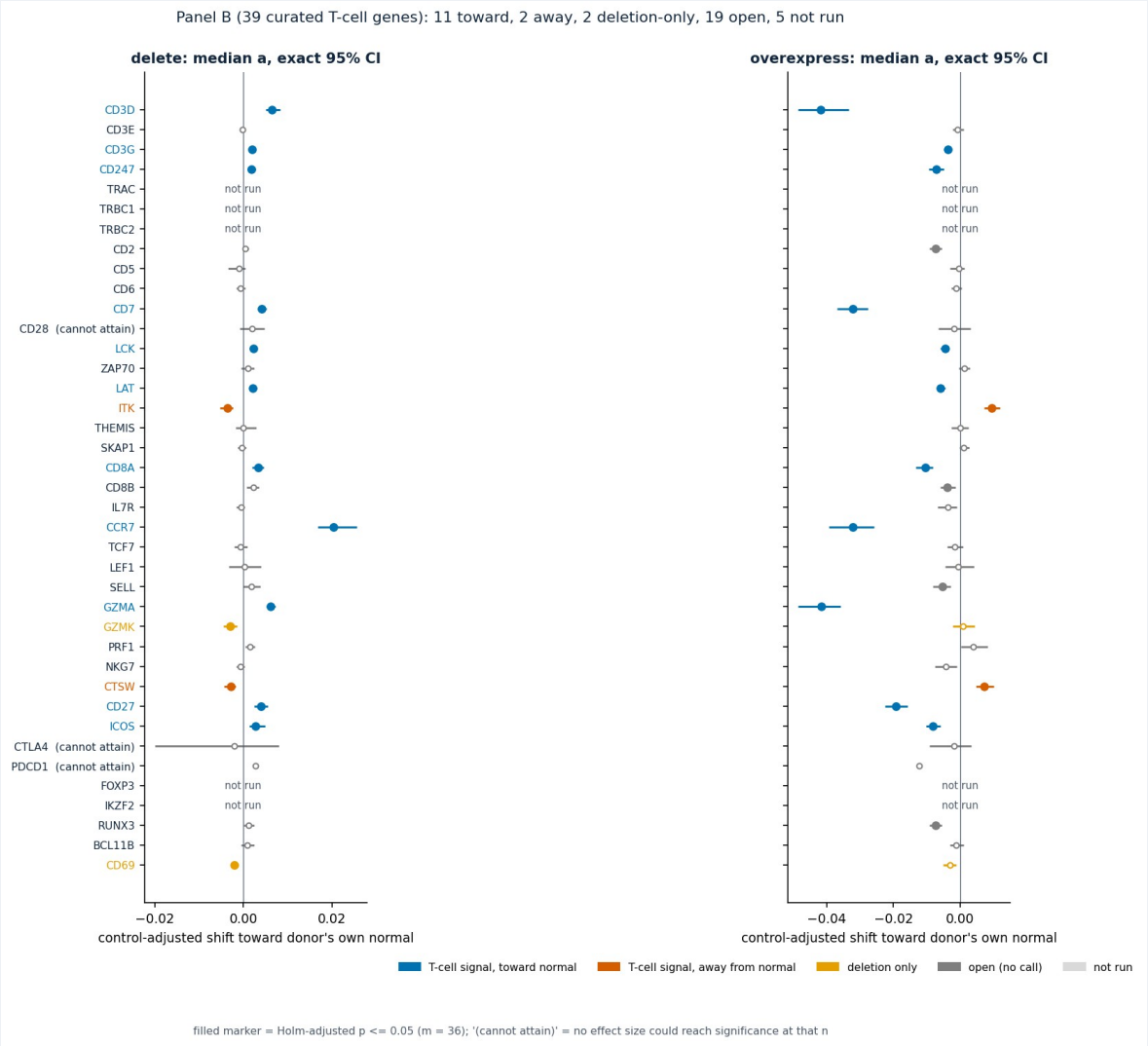
Keratins, mucin, a secretoglobin, a haemoglobin chain: epithelial and blood markers, present in ≤ 8 of 43 donors' T cells.

The one testable gene, POLR2J3, is OPEN ($p = 0.10 / 0.25$).

The design did not refute the July panel; it showed the panel is mostly not T-cell biology.

Confound: 316M replaced 104M (no 104M arm), so a non-replication could not be attributed — moot, none occurred.

13 of 34 tested T-cell genes give a dose-concordant signal



Toward normal (11)

CD3D, CD3G, CD247, CD7, LCK, LAT, CD8A, CCR7, GZMA, CD27, ICOS

Away from normal (2)

ITK, CTSW

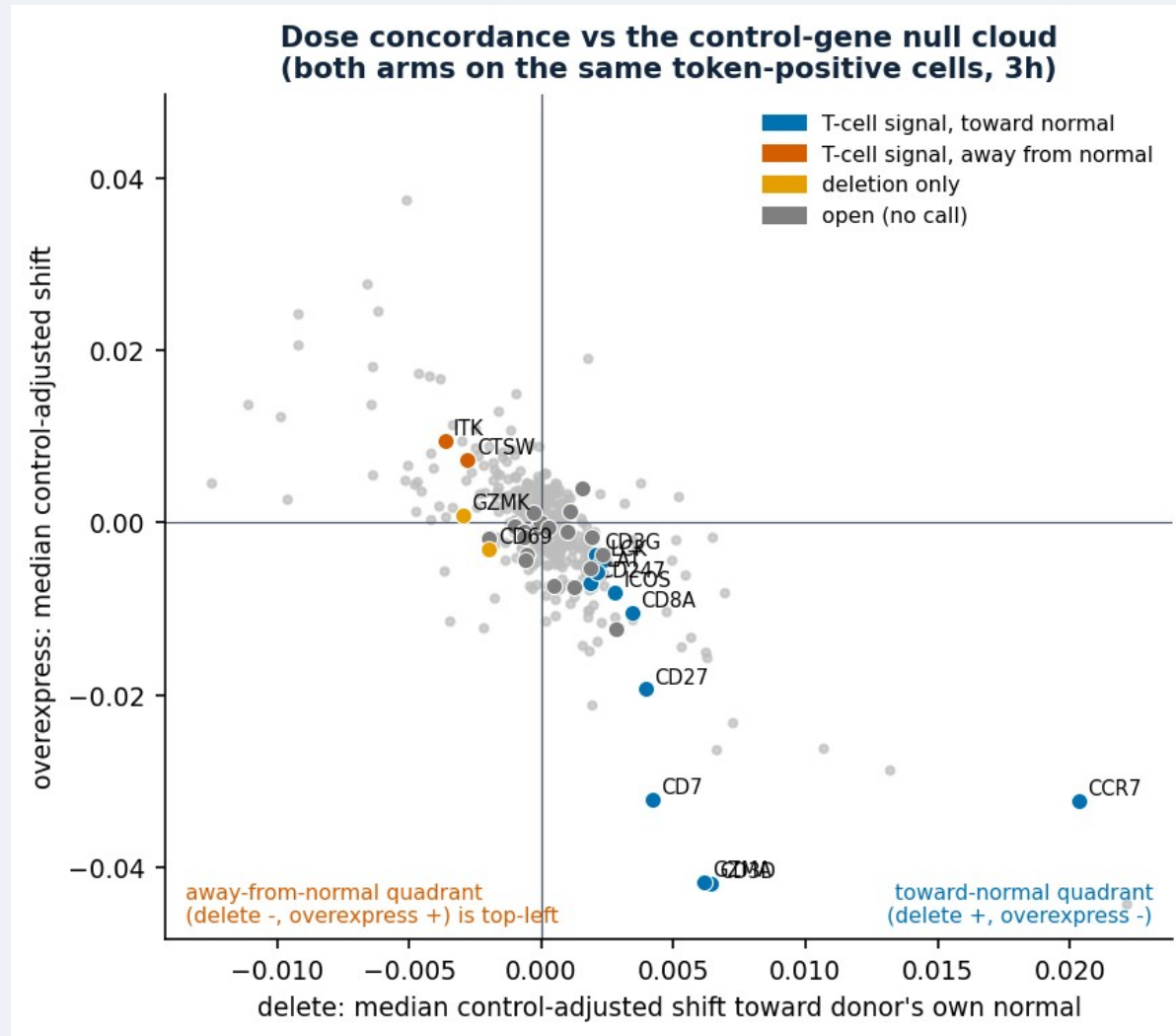
Stable — and what cannot be judged

None of the 13 moves under any registered sensitivity.

Below the design $d_{\min}$ of 11 (registered): CD28, CTLA4, LEF1, PDCD1. Of these, CD28, CTLA4, PDCD1 cannot attain significance at any effect size in this family (post hoc) — no evidence either way.

TRAC/TRBC1/TRBC2: not in the model's vocabulary.

Opposite-signed arms are common even for control genes — effect size separates few



Confound

360 matched control entries are anti-correlated too ($\rho = -0.62$); 35% sit in the toward-normal quadrant.

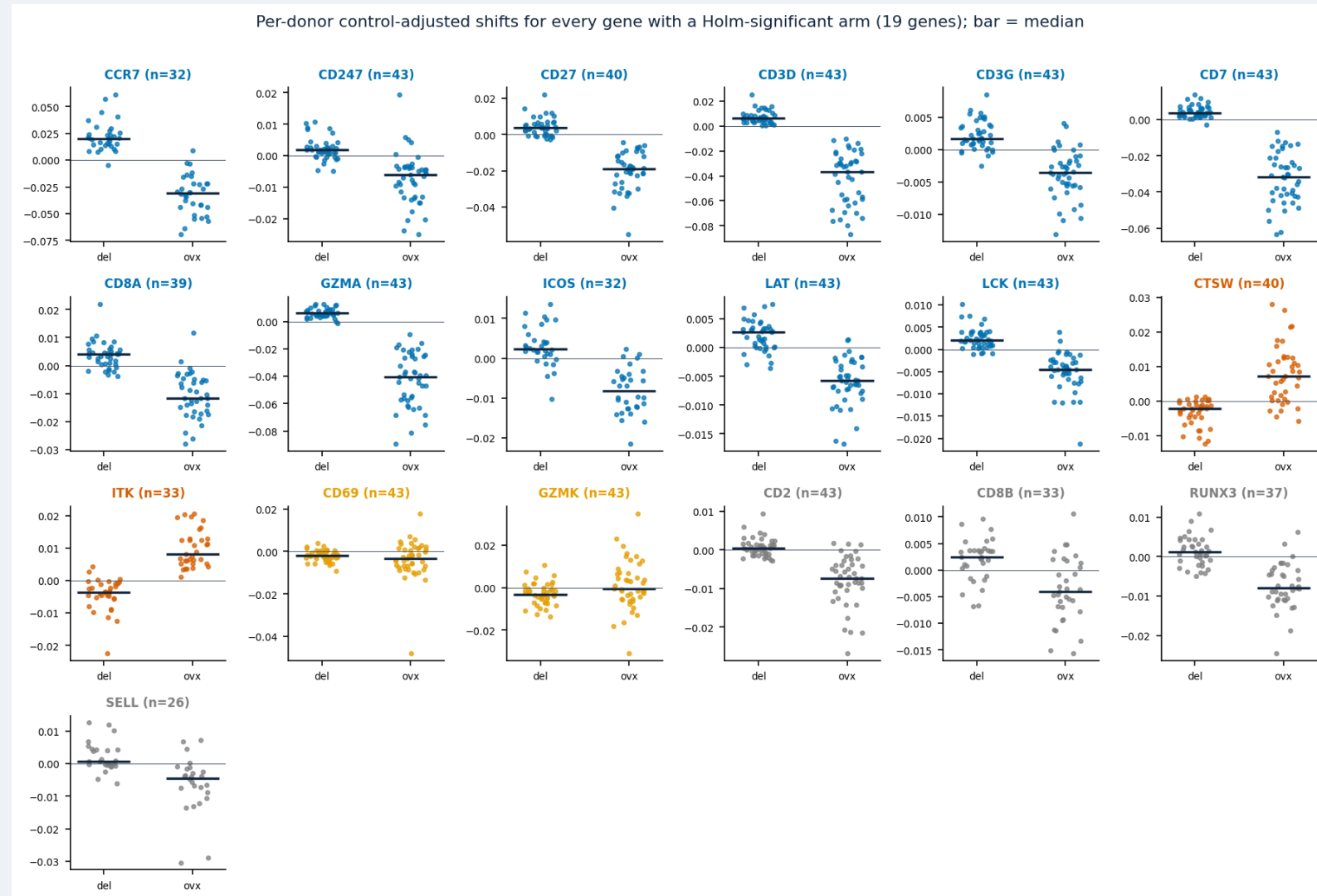
Post hoc, descriptive: only CCR7, CD3D, GZMA, CD7 (and CD27 on overexpression) lie beyond ~95% of controls.

Measured on the same cells (Amendment 3h)

Geneformer overexpresses into all cells; the token-positive subset was reconstructed and checked.

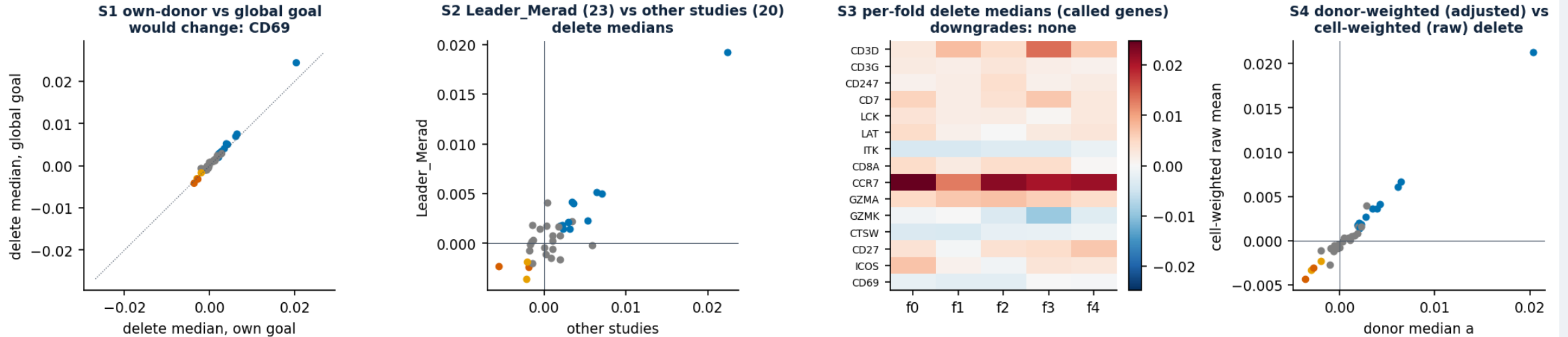
15,142 / 15,142 count checks; 8 / 8 GPU identity pairs bit-identical.

The medians are not carried by a few donors



The coherent set does not move under any registered sensitivity

Sensitivities: reported alongside, never used to change a registered status



Three fragile rows, outside the coherent set

GZMK: deletion-only → away under all-cells overexpress. PRF1: open → deletion-only if any control counts. CD69: deletion-only → dose-incoherent under Holm-over-tested and under the global goal.

Panel-level: 21 toward vs 13 away, sign test $p = 0.23$ — genes share controls, so signs are not independent.

What each status licenses — and what it does not

T-cell signal (13)

Licenses: Model-internal, donor-consistent, dose-concordant response beyond matched controls.

Does not: Not a causal driver; mostly not unusual among genes.

Deletion only (2)

Licenses: A consistent deletion effect.

Does not: No direction; no signal claim.

Open (20)

Licenses: The registered test did not call it.

Does not: Not 'no effect'; for CD28/CTLA4/PDCD1 no evidence either way.

Not run (18) • no controls (1)

Licenses: Not expressed in enough donors' T cells, or not in the vocabulary.

Does not: Not non-replication; not no effect.

What this design still cannot rule out

Ambient RNA

Tumour vs normal tissue differ in contamination within a donor; pairing cannot remove it.

Model change

316M bf16 replaced July's 104M fp32; no 104M arm.

Not independent

20 of 43 donors overlap July's LUAD pool.

Selection

T-cell-rich samples only; 17 unpaired donors excluded.

Nondeterminism

Fine-tuning is not bit-reproducible; the ISP inherits these fold models.

Process

No-op and wrapper checks ran after Phase 6 started; the overexpress cell set was fixed by Amendment 3h after Phase 6.

For the July screen: its LUAD → normal list was mostly not T-cell biology

What this changes

13 of 15 July hits cannot be tested in T cells under a paired, balanced design.

The model does read canonical T-cell genes (CD3 complex, LCK, LAT, CD8A, CCR7, GZMA) in a donor-consistent, dose-concordant way.

Effect size, not significance, separates a handful from arbitrary genes.

Next

Genome-wide control distribution, to rank effect size formally (registered first).

Checkpoint genes (CTLA4, PDCD1, CD28) need more token-positive donors: a larger cohort or a looser cap.

A 104M arm, if attribution to model vs design is wanted.

Independent tissue validation for the top four (CCR7, CD3D, GZMA, CD7).